

sTMS Interface Functional Specification

Institute of Technical Education – Service
Indicator

Version 1.0

Document Version

Version	Effective Date	Summary of Changes	Created/ Updated By
0.1	22/03/2022	The initial version of sTMS - ITE Service Indicator interface functional specification.	AvePoint
0.2	23/05/2022	Revised version of sTMS – ITE Service Indicator interface functional specification based on feedback • Update on 2. summary of the design interface • 3. Interface field - Included 4 tables, such as External Organization Table, Audit table, Service indicator for Individuals and Organizations • Update on 4. Assumption and Constraints • 5. Appendix - Remove SS2 from Mapping reasoning from ITE to sTMS	AvePoint
0.3	01/06/2022	Updated Table 3.1 External Organization Table and respective fields based on client's feedback	AvePoint
1.0	03/06/2022	Finalized Version by ITE and AvePoint	AvePoint

Document Acceptance Form

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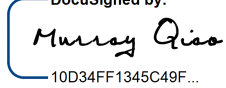
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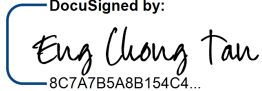
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1. Introduction

1.1 Purpose of this document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirements and detailed technical specification between the SaaS shared Training Management System (sTMS) and ITE Service Indicator for student suspended list records.

sTMS and ITE Service Indicator shall exchange student suspended list records by generating corresponding CSV files encrypted by GnuPG in the SFTP server accordingly.

1.2 Intended Audience

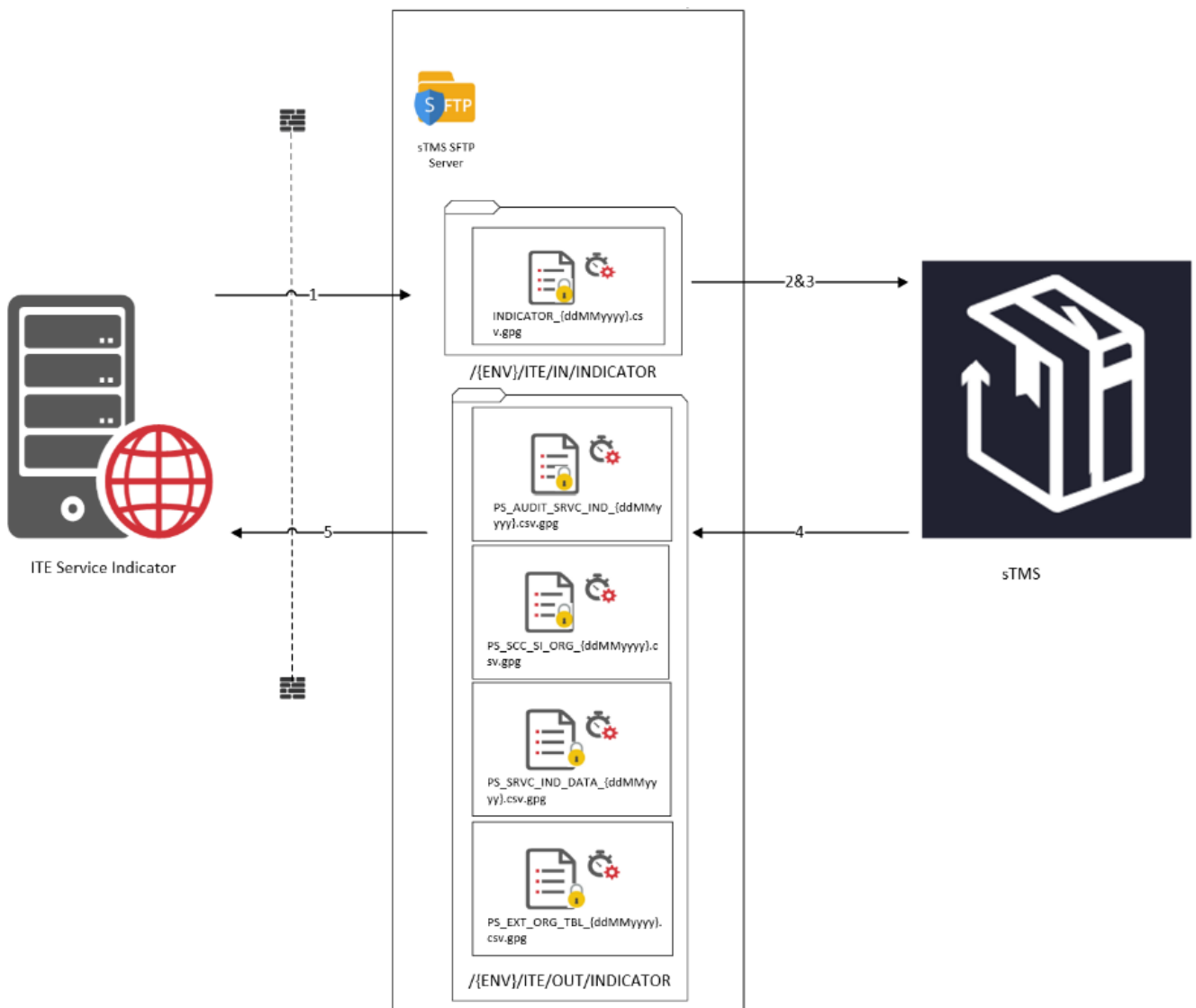
This technical document is used for system-to-system communication. Therefore, the intended audiences are the technical stakeholders of both systems.

2. Technical Overview

2.1 System Flow

There are 3 roles involved in the interface diagram.

- **ITE Service Indicator:** Institute of Technical Education Service Indicator System.
- **sTMS SFTP Server:** A SFTP server hosted in Azure that stores encrypted CSV files used for data exchange.
- **sTMS:** shared Training Management System.



SN	Description
1	ITE Service Indicator System generates the PET student suspended records and pushes the file into the sTMS SFTP server after encryption with GnuPG.
2	sTMS pull the feed file on a pre-defined schedule and retrieve the records after decryption. The records would be saved to sTMS.
3	sTMS generates the CET student suspended records into 4 different CSV files based on the list of external organizations table, service indicator for organization and students and audit records on a pre-defined schedule.
4	sTMS encrypts the feed file with GnuPG and push the encrypted file into sTMS SFTP server on a predefined schedule.
5	ITE pulls the interface files on a pre-defined schedule and retrieves the records after decryption.

2.2 Interface Design Summary

Input Interface File

Interface File Name	INDICATOR_{ddMMyyyy}.CSV.gpg
Direction	ITE Service Indicator to sTMS
Mode of Interface	CSV file
Method	SFTP
File Path	/{ENV}/ITE/IN/INDICATOR ENV: UAT, PROD
Encryption Method	PGP with GnuPG
Data Syncing Mode	First-time full data sync + Subsequent delta data sync
Frequency	Daily
Timing	08:00pm
File Transfer Policy	Empty CSV file with header info shall be generated and pushed into sTMS SFTP if there are new no records from ITE Service Indicator.
Possible Exception and Exception Handling	File Transfer failure due to file not found or invalid data field. Exception handling steps:

	• ITE Service Indicator vendor shall be informed of the job failure reason by AP • ITE Service Indicator vendor to troubleshoot the cause of the failure. • If the issue is due to the transfer files could not be found in SFTP: ○ ITE redeposits the files in sTMS SFTP. ○ AP to trigger the transfer job.
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Output Interface File

Interface File Name	PS_AUDIT_SRVC_IND_{ddMMyyyy}.CSV.gpg PS_SCC_SI_ORG_{ddMMyyyy}.CSV.gpg PS_SRVC_IND_DATA_{ddMMyyyy}.CSV.gpg PS_EXT_ORG_TBL_{ddMMyyyy}.CSV.gpg
Direction	sTMS to ITE Service Indicator
Mode of Interface	CSV file
Method	SFTP
File Path	/{ENV}/ITE/OUT/INDICATOR ENV: UAT, PROD
Encryption Method	PGP with GnuPG
Data Syncing Mode	First-time full data sync + Subsequent delta data sync Filter by the institution which refers to "ITE"
Frequency	Daily
Timing	09:00pm
File Retention Policy	Files are retained for a maximum of 30 days in the SFTP server, after this period the files will be automatically deleted. ITE can delete the files after pulling out
File Transfer Policy	Empty CSV file with header info shall be generated and pushed to sTMS SFTP if there are new no records.
Possible Exception and Exception Handling	File transfer failure due to sTMS internal exception such as timer service down. Exception handling steps: • sTMS support team shall be informed by email of the transfer failure. • AP team checks the relevant logs to troubleshoot the cause of the failure. • If the issue is due to the transfer files could not be found in SFTP, AP redeposits the files in SFTP

2.3 Business Rule

No.	Business Rule
1	CET student would be eligible to join PET course if student does not have any outstanding fee in ITE iStudent and negative indicators in Service Indicator.
2	sTMS will block the application if student/organization is found with bad debts in PET courses from the ITE suspended records.
3	Students who are classified under Special Service Indicator 2 (SS2) are still eligible for CET courses.

3. Interface Fields

3.1 External Organizations Table

The import process will check if the data already exists in iStudent based on the EXT_ORG_ID. The process should not update if the EXT_ORG_ID already exists in iStudent. In the event that the EXT_ORG_ID does not exist, the system will insert the new row of data into iStudent.

Name	Description	Remarks
EXT_ORG_ID	Organization UEN number	UEN of company
EFFDT	Effective Date	Always '01/01/1970'
EFF_STATUS	Effective Status	Always 'A'
OTH_NAME_SORT_SRCH	Search Name	Nil
EXT_ORG_TYPE	Organization Type	Always 'OTHR'
DESCR	Company Name	Format: String
DESCR50	Company Name	Format: String
DESCRSHORT	Company Name	Format: String
ORG_CONTACT	Contact Number	Default to 0
ORG_LOCATION	Location Number	Default to 0
ORG_DEPARTMENT	Department Number	Default to 0
PROPRIETORSHIP	Proprietorship	Always 'OTHR'
SETID	SetID	Always 'ITESA'
VENDOR_ID	Vendor ID	Nil
TAXPAYER_ID_NO	Taxpayer ID Number	Nil
LASTUPDDTTM	Last Update Date/Time	Default to System Date Time Format: <i>dd-MMM-yy hh.mm.ss.ffffff00tt</i>
LASTUPDOPRID	By	Default to Workday User ID of Process Run 8 digital. Format: 10002841

3.2 Service Indicator for Organizations

Based on the ACTN code, the process will insert/update or delete the row of data in iStudent.

Name	Description	Values
ACTN	Action code.	C: Create U: Update D: Delete
EXT_ORG_ID	Organization UEN number	UEN of company
SRVC_IND_DTTM	Service Indicator Date Time	Format: <i>dd-MMM-yy hh.mm.ss.ffffff00tt</i>
OPRID	The ITE staff who create/update student suspended records.	ID from Workday HR. 8 digital. Format: 10002841
INSTITUTION	Institution Name.	Default to 'ITE'
SRVC_IND_CD	Service indicator code.	Please refer to the Reason description table in Appendix
SRVC_IND_REASON	Service indicator reason.	Please refer to the Reason description table in Appendix
SRVC_IND_ACT_TERM	Service indicator action term. Not used in sTMS, leave as blank.	Nil
SCC_SI_END_TERM	Service indicator end term. Not used in sTMS, leave as blank.	Nil
SCC_SI_END_DT	Effective to.	Format: <i>dd-MMM-yy</i>
SRVC_IND_ACTIVE_DT	Date of effective from.	Format: <i>dd-MMM-yy</i>
POS_SRVC_INDICATOR	Positive service indicator.	Always 'N'
SRVC_IND_REFRNCE	Service indicator reference. Not used in sTMS, leave as blank.	Nil
DEPTID	Department id.	Always 'ITE'
POSITION_NBR	NA	Nil
CONTACT	NA	Nil
CONTACT_ID	NA	Nil
CURRENCY_CD	NA	Nil else Default to SGD if AMOUNT is not 0
AMOUNT	The total amount that an individual or company owes the institution.	Format: Number
SEQ_3C	NA	Nil
PLACED_METHOD	Not used in sTMS, default to 'M'.	Always 'M'

PLACED_PERSON	Placed person, person name of created by.	Format: String
PLACED_PERSON_ID	Placed person ID from Workday HR system.	ID from Workday HR. 8 digital. Format: 10002841
PLACED_PROCESS	Not used in sTMS, default to 'sTMS'.	sTMS
RELEASE_PROCESS	NA	Nil
COMM_COMMENTS	Service indicator comments.	Format: String

3.3 Service Indicator for Individual Data

Based on the ACTN code, the process will insert/update or delete the row of data in iStudent. Use COMMON_ID to check against PS_EXTERNAL_SYSTEM. EXTERNAL_SYSTEM_ID to get the EMPLID of the student.

Name	Description	Values
ACTN	Action code.	C: Create U: Update D: Delete
COMMON_ID	UAM common id.	
EMPLID	For Singaporeans, EP, SP holder, shall be NRIC/FIN, for the foreigner without FIN shall be Passport number.	student NRIC/FIN/Passport
EXT_ORG_ID	Organization UEN number	UEN of company
SRVC_IND_DTTM	Service Indicator Date Time	Format: <i>dd-MMM-yy hh.mm.ss.ffffff00 tt</i>
OPRID	The ITE staff who create/update student suspended records.	ID from Workday HR. 8 digital. Format: 10002841
INSTITUTION	Institution Name.	Default to 'ITE'
SRVC_IND_CD	Service indicator code.	Please refer to the Reason description table in Appendix
SRVC_IND_REASON	Service indicator reason.	Please refer to the Reason description table in Appendix
SRVC_IND_ACT_TERM	Service indicator action term. Not used in sTMS, leave as blank.	Nil
SCC_SI_END_TERM	Service indicator end term. Not used in sTMS, leave as blank.	Nil
SCC_SI_END_DT	Effective to.	Format: <i>dd-MMM-yy</i>
SRVC_IND_ACTIVE_DT	Date of effective from.	Format: <i>dd-MMM-yy</i>

POS_SRVC_INDICATOR	Positive service indicator.	Always 'N'
SRVC_IND_REFRNCE	Service indicator reference. Not used in sTMS, leave as blank.	Nil
DEPTID	Department id.	Always 'ITE'
POSITION_NBR	NA	Nil
CONTACT	NA	Nil
CONTACT_ID	NA	Nil
CURRENCY_CD	NA	Nil else Default to SGD if AMOUNT is not 0
AMOUNT	The total amount that an individual or company owes the institution.	Format: Number
SEQ_3C	NA	Nil
PLACED_METHOD	Not used in sTMS, default to 'M'.	Always 'M'
PLACED_PERSON	Placed person, person name of created by.	Format: String
PLACED_PERSON_ID	Placed person ID from Workday HR system.	ID from Workday HR. 8 digital. Format: 10002841
PLACED_PROCESS	Not used in sTMS, default to 'sTMS'.	sTMS
RELEASE_PROCESS	NA	Nil
COMM_COMMENTS	Service indicator comments.	Format: String

3.4 Audit Records for Service Indicator Data

Name	Description	Values
AUDIT_OPRID	The ITE staff who create/update student suspended records. Same with OPRID.	ID from Workday HR. 8 digital. Format: 10002841
AUDIT_STAMP	Student suspended record modification timestamp.	Format: <i>dd-MMM-yy</i> <i>hh.mm.ss.ffffff00 tt</i>
AUDIT_ACTN	Action code.	AC: Add/Create C: Change/Update D: Delete
COMMON_ID	UAM common id.	
EMPLID	For Singaporeans, EP, SP holder, shall be NRIC/FIN, for the foreigner without FIN shall be Passport number.	student NRIC/FIN/Passport

EXT_ORG_ID	Organization UEN number	UEN of company
SRVC_IND_DTTM	Service Indicator Modified Time in sTMS	Format: <i>dd-MMM-yy hh.mm.ss.ffffff00 tt</i>
OPRID	The ITE staff who create/update student suspended records.	ID from Workday HR. 8 digital. Format: 10002841
INSTITUTION	Institution Name.	Default to 'ITE'
SRVC_IND_CD	Service indicator code.	Please refer to the Reason description table in Appendix
SRVC_IND_REASON	Service indicator reason.	Please refer to the Reason description table in Appendix
SRVC_IND_ACT_TERM	Service indicator action term. Not used in sTMS, leave as blank.	Nil
SCC_SI_END_TERM	Service indicator end term. Not used in sTMS, leave as blank.	Nil
SCC_SI_END_DT	Effective to.	Format: <i>dd-MMM-yy</i>
SRVC_IND_ACTIVE_DT	Date of effective from.	Format: <i>dd-MMM-yy</i>
POS_SRVC_INDICATOR	Positive service indicator.	Always 'N'
SRVC_IND_REFRNCE	Service indicator reference. Not used in sTMS, leave as blank.	Nil
DEPTID	Department id.	Always 'ITE'
POSITION_NBR	NA	Nil
CONTACT	NA	Nil
CONTACT_ID	NA	Nil
CURRENCY_CD	NA	Nil
AMOUNT	The total amount that an individual or company owes the institution.	Format: Number
SEQ_3C	NA	Nil
PLACED_METHOD	Not used in sTMS, default to 'M'.	Always 'M'
PLACED_PERSON	Placed person, person name of created by.	Format: String
PLACED_PERSON_ID	Placed person ID from Workday HR system.	ID from Workday HR. 8 digital. Format: 10002841
PLACED_PROCESS	Not used in sTMS, default to 'sTMS'.	sTMS
RELEASE_PROCESS	NA	Nil
COMM_COMMENTS	Service indicator comments.	Format: String

3.5 ITE to sTMS Service Indicator Fields

Name	Description	Values
AUDIT_OPRID	The ITE staff who create/update student suspended records.	ID from Workday HR. 8 digital. Format: 10002841
AUDIT_STAMP	Student suspended record modification timestamp.	Format: <i>dd-MMM-yy</i> <i>hh.mm.ss.ffffff00 tt</i>
AUDIT_ACTN	Action code.	A: Add/Create C: Chang/Update D: Delete
EMPLID	For Singaporeans, EP, SP holder, shall be NRIC/FIN, for the foreigner without FIN shall be Passport number.	student NRIC/FIN/Passport
EXT_ORG_ID	Organization UEN number	UEN of company
SRVC_IND_DTTM	Service Indicator data based on Audit Records	Format: <i>dd-MMM-yy</i> <i>hh.mm.ss.ffffff00 tt</i>
OPRID	The ITE staff who create/update student suspended records.	ID from Workday HR. 8 digital. Format: 10002841
INSTITUTION	Institution Name.	Default to 'ITE'
SRVC_IND_CD	Service indicator code.	Please refer to the Reason description table in Appendix
SRVC_IND_REASON	Service indicator reason.	Please refer to the Reason description table in Appendix
SRVC_IND_ACT_TERM	Service indicator action term. sTMS can only support SRVC_IND_DTTM, if SRVC_IND_DTTM is empty, ITE needs to convert SRVC_IND_ACT_TERM to DateTime and send the DateTime by SRVC_IND_DTTM.	Nil
SCC_SI_END_TERM	Service indicator end term. sTMS can only support SCC_SI_END_DT, if SCC_SI_END_DT is empty, ITE needs to convert SCC_SI_END_TERM to DateTime and send the DateTime by SCC_SI_END_DT.	Nil
SCC_SI_END_DT	Effective to.	Format: <i>dd-MMM-yy</i>
SRVC_IND_ACTIVE_DT	Date of effective from.	Format: <i>dd-MMM-yy</i>
POS_SRVC_INDICATOR	Positive service indicator.	Always 'N'
SRVC_IND_REFRNCE	Service indicator reference.	Nil
DEPTID	Department id.	Always 'ITE'

POSITION_NBR	NA	Nil
CONTACT	NA	Nil
CONTACT_ID	NA	Nil
CURRENCY_CD	NA	Nil
AMOUNT	The total amount that an individual or company owes the institution.	Format: Number
SEQ_3C	NA	Nil
PLACED_METHOD	NA	Always 'M'
PLACED_PERSON	Placed person, person name of created by.	Format: String
PLACED_PERSON_ID	Placed person ID from Workday HR system.	ID from Workday HR. 8 digital. Format: 10002841
PLACED_PROCESS	NA	Nil
RELEASE_PROCESS	NA	Nil
COMM_COMMENTS	Service indicator comments.	

4. Constraints

No.	Constraints
1	sTMS will only send MOE suspended list and Other suspended list to ITE Service Indicator System.

5. Appendix

5.1 Reason Mapping for sTMS to ITE

Reason in sTMS	Mapping to SRVC_IND_CD in ITE	Mapping to SRVC_IND_REASON in ITE
Finance related	DEB	DEBTS
Others	CSI	OTHER
Finance related to MOE Suspended List	MOE	MOE

5.2 Reason Mapping for ITE to sTMS

SRVC_IND_CD	SRVC_IND_REASON	SRVC_IND_REASON_DESCR	Mapping to sTMS
ADJ	ADJST	Adjustment of Fees	Finance related
BAD	SMOKE	Caught Smoking	Others
	OTHER	Other Reasons	Others
	CRIME	Committed Criminal Offence	Others
	FIGHT	Caught Fighting	Others
BRB	BRADM	Break Bond for AOM	Finance related
CSI	FIGHT	Caught Fighting	Others
	SMOKE	Caught Smoking	Others
	CRIME	Committed Criminal Offence	Others
	OTHER	Other Reasons	Others
DBM	MISCD	Misconduct	Others
	BWDAB	BREAK WDA SPONSORSHIP BOND	Finance related
	ADMIS	Admission	Others
DBR	ATTEN	Poor Attendance	Others

	OTHER	Other Reasons	Others
DEB	DEBTS	Bad Debts	Finance related
DFR	NS	National Service	Others
	OTHER	Other Reasons	Others
	OVRSE	On Overseas Assignment	Others
	MEDIC	Medical Reasons	Others
	HOSP	Hospitalised	Others
	RESVR	Reservist Training	Others
DID	DID	Duplicate Identification Docum	Others
IDC	IDHBC	ID has been Changed	Others
INS	INSTA	Payment by Instalment	Finance related
INT	INTH	Interview-Health Reason	Others
	INTD	Interview-Disciplinary Reason	Others
MCD	EXMCD	Exceeded Max Cand Period	Others
MOE	MOE	Owe MOE Fees	Finance related to MOE Suspended List
MPA	MPADM	MP Admission	Others
RFD	NORFD	Hold Refund Amount	Finance related
RMK	ADRMK	Additional Remarks	Others
SSI	DA2YR	Debar Admissions for 2 years	Others
WEC	OSFEE	Withhold eCert for Owe ITE Fee	Finance related

5.3 Sample files

    
 INDICATOR_250520 PS_AUDIT_SRVC_IND PS_EXT_ORG_TBL_24 PS_SCC_SI_ORG_240 PS_SRVC_IND_DATA
 22.csv _24052022.csv 052022.csv 52022.csv _24052022.csv